

Measurement - Mass and Volume Towards Density

DATA

Table 1: Measurements of Regular 3D Shapes

	Mass (g)	a (mm)	b (mm)	c (mm)
(1) Cube	15.875	29.50	30.00	29.20
(2) Cuboid	31.380	60.35	29.50	29.85
(3) Triangular Prism	13.672	60.50	28.30	29.05
(4) Cylinder	21.250	60.25	29.40	28.95
(5) Square Pyramid	11.685	59.95	29.15	28.65
(6) Triangular Pyramid	4.444	60.25	27.00	27.40
(7) Cone	10.085	55.25	29.15	29.25
(8) Ovoid	16.171	_____	_____	_____
(9) Ellipsoid	17.249	54.95	29.30	29.65
(10) Sphere	8.767	29.15	30.00	29.95

YOUR PROCEDURE & OBSERVATIONS

We calibrated the scale, then measured the mass of each 3D shape individually. We then both measured different shapes in order to save time. The curved and triangular shapes were difficult to accurately measure. For the curved shapes, we tried to measure the longest part, and the diameters orthogonal to it. For triangular shapes, we measured the length, and the width and height of the triangle.

Table 2: Measurements of Fluorite

	Mass (g)	Volume (mL)	Volume of Water ↓
1	1.204	0.4	4.9
2	0.918	0.3	7.0
3	1.212	0.3	6.1
4	1.754	0.5	4.4
5	1.185	0.4	6.5
6	0.745	0.2	6.5
7	0.844	0.2	3.7
8	1.626	0.5	4.7
9	0.423	0.1	7.2
10	0.987	0.3	7.3

	Mass (g)	Volume (mL)	Volume of Water ↓
11	0.921	0.3	5.4
12	1.267	0.4	6.5
13	1.023	0.3	5.6
14	0.734	0.2	6.9
15	0.509	0.2	7.5
16	0.652	0.2	4.2
17	0.542	0.2	8.2
18	0.762	0.2	5.1
19	1.265	0.4	2.6
20	0.868	0.3	3.5

Dr. Virginia Chang

Volume of Water

YOUR PROCEDURE & OBSERVATIONS

We measured each fluorite by mass and volume one at a time. When measuring volume, we measured the volume of the water before placing in the fluorite, and subtracted it from the volume of the water and the fluorite to get the volume of the fluorite. While the volume was similar between most of the fluorite, the mass often changed.

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